

Spam Message Classification using Machine Learning

Project Documentation

Project Overview

This project develops a spam message classification system using supervised machine learning algorithms to classify messages as Spam or Legitimate.

Objectives

Build a spam classifier, preprocess data, apply feature scaling, train KNN, SVM, and Gaussian Naive Bayes models, and compare their performance.

Dataset

The dataset contains around 5,200 messages with message features and the target variable spam_label.

Workflow

Data Loading → Data Exploration → Feature Scaling → Train-Test Split → KNN → SVM → Naive Bayes → Model Evaluation → Comparison.

Algorithms Used

K-Nearest Neighbors (KNN), Support Vector Machine (Linear & RBF), Gaussian Naive Bayes.

Evaluation Metrics

Accuracy, Precision, Recall, F1 Score, Confusion Matrix, ROC Curve, and ROC-AUC.

Visualizations

Spam distribution, correlation heatmap, confusion matrices, ROC curves, and model comparison charts.

Conclusion

Among the implemented models, the SVM with the RBF kernel generally achieved the best overall performance for spam message classification, while Naive Bayes provided fast predictions and KNN served as a reliable baseline.

Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Google Colab.